

Evaluating Physics-Informed Neural Networks as a Framework for Scientific Computing

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Implement, analyze, and evaluate Physics-Informed Neural Networks (PINNs) as a framework for modeling physical systems governed by PDEs.

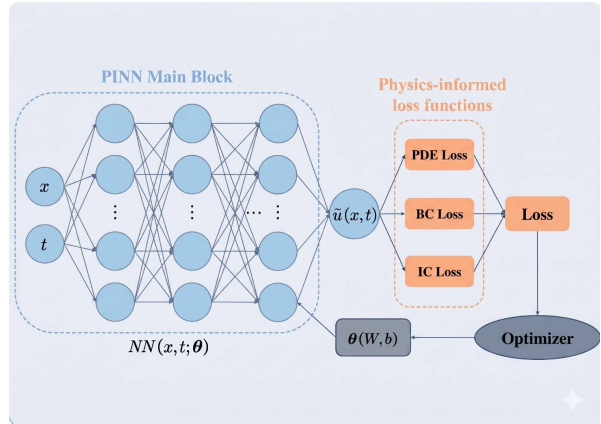
Evaluate whether PINNs can replace traditional numerical methods such as Explicit Finite Difference Method (FDM) in scientific computing.

Main Comparison Criteria

- Stability
- Accuracy
- Computational Efficiency
- Physical Consistency
- Boundary Flexibility

Processing Workflow

- 1. Input:** Spatial (x, y, z) and temporal (t) coordinates are fed into the neural network.
- 2. Forward Pass:** The network predicts the solution $u(x, t)$.
- 3. Differentiation:** Automatic Differentiation (AD) computes derivatives with respect to input coordinates.
- 4. Evaluation:** The predicted derivatives are substituted into the governing PDE to compute residuals.



The Finite Difference Method (FDM) discretizes the domain into a structured grid. The explicit scheme advances the solution in time based on previous steps.

$$u_i^{n+1} = u_i^n + \Delta t \cdot f(u_i^n, \nabla u_i^n)$$

Limitations:

Highly sensitive to the CFL stability condition. If Δt is too large, the simulation diverges.

Key FDM Traits

- Localized computations (stencils)
- Convergence depends on grid density
- Difficult to apply to irregular geometries

1. Solve Forward Problems

- PINN and FDM for 1D diffusion
- Extend to 2D bounded diffusion
- PINN for 1D Nonlinear Schrödinger Equation

2. Solve Inverse Problems

- Dual-network PINN to identify hidden physical defects from noisy sensor data

3. Comparative Analysis

- Accuracy
- Stability
- Computational Efficiency
- Scalability
- Boundary Handling

PINN for the Nonlinear Schrödinger Equation

Governing Equation

$$ih_t + \frac{1}{2}h_{xx} + |h|^2h = 0$$

$$x \in [-5, 5], \quad t \in [0, \pi/2]$$

Initial Condition

$$h(0, x) = 2 \operatorname{sech}(x)$$

Architecture

- 5 hidden layers
- 100 neurons per layer
- tanh activation

Key Results

- Stable soliton-like structure
- Symmetric peak evolution
- Smooth oscillation-free profile

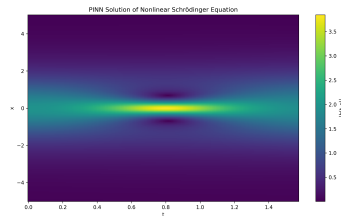


Fig. 1: Spatio-temporal evolution

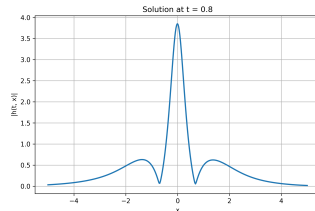


Fig. 2: Time slice at $t = 0.8$

Governing Equation

$$u_t = Du_{xx}$$

Initial Condition: Gaussian pulse

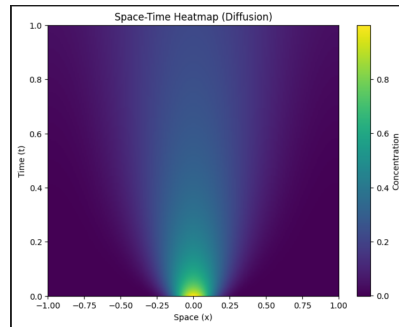
Boundary Condition: Reflective Neumann BC

Architecture

- 4 hidden layers
- 64 neurons
- tanh activation
- Adam optimizer

PINN Characteristics

- Continuous mesh-free solution
- No CFL stability restriction
- Physics enforced through PDE residual minimization



Heatmap for 1-D diffusion showing spreading Gaussian profile.

FTCS Scheme

$$u_i^{n+1} = u_i^n + r(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

where

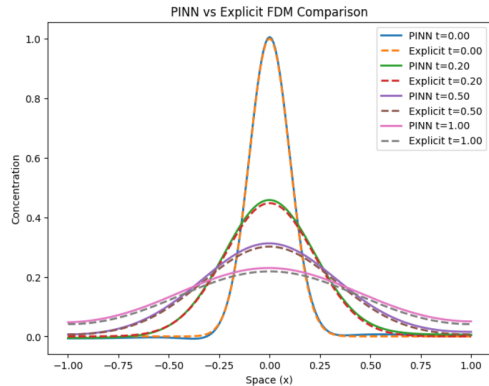
$$r = \frac{D\Delta t}{\Delta x^2}$$

CFL Stability Condition

$$r \leq 0.5$$

FDM Characteristics

- Deterministic numerical method
- Computationally efficient
- Grid-based discretization
- Difficult for irregular geometries



Property	PINN	Explicit FDM
Stability	No CFL constraint	CFL required
Output	Continuous	Grid-based
Inverse Problems	Supported	Not supported
Scalability	Mesh-free	Grid explosion

Asymmetric Boundary Diffusion (After Midsem)

System Setup

$$x \in [-5, 25], \quad t \in [0, 900], \quad D = 0.1$$

Left BC: Absorbing

Right BC: Reflective

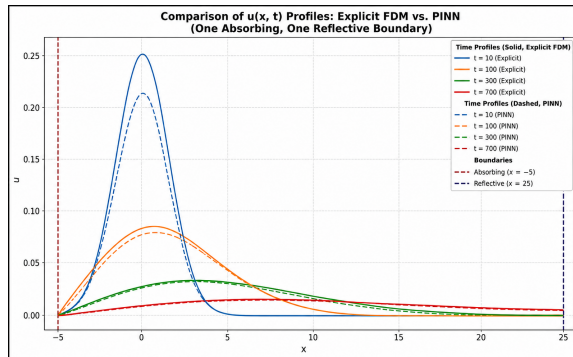
Loss Function

$$L_{\text{Total}} = L_{\text{PDE}} + \lambda_{BC}(L_{BC, \text{left}} + L_{BC, \text{right}}) + \lambda_{IC} L_{IC}$$

Important Training Strategies

- Hybrid collocation sampling near Gaussian peak
- Initial-condition pretraining for stability
- Adam optimization followed by L-BFGS refinement

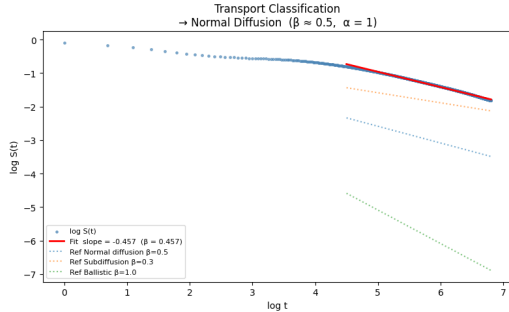
Computational Observation: PINNs remove CFL restrictions but require expensive multi-phase optimization. FDM performance is affected by asymmetric boundaries, due to the need of ghost-node generation.



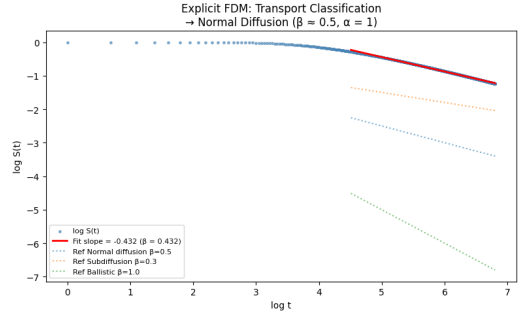
PINN (dashed) vs FDM (solid) at multiple time instances.

Transport Classification

Transport scaling relation: $S(t) \propto t^{-\beta}$, Normal diffusion predicts $\beta \approx 0.5$



$$\beta_{\text{PINN}} = 0.457$$



$$\beta_{\text{FDM}} = 0.432$$

Both frameworks classify the system as **Normal Diffusion**.

PINN produces a transport exponent closer to the theoretical prediction.

Two Absorbing Boundaries

System Setup:

$$u_t = Du_{xx}, \quad D = 0.1, \quad x \in [-5, 25], \quad t \in [0, 900]$$

Boundary Conditions

$$u(t, -5) = 0, \quad u(t, 25) = 0$$

Both boundaries are perfectly absorbing.

Loss Function

$$L_{\text{Total}} = L_{\text{PDE}} + \lambda_{BC}(L_{BC, \text{left}} + L_{BC, \text{right}}) + \lambda_{IC}L_{IC}$$

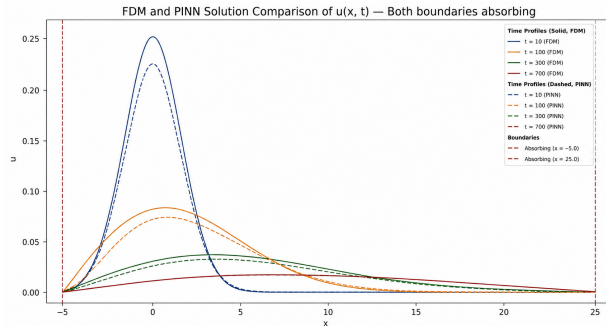
Transport Classification

$$MSD(t) \propto t^\alpha$$

Normal diffusion predicts: $\alpha \approx 1$

Results

Both frameworks reproduce the expected exponential survival decay.



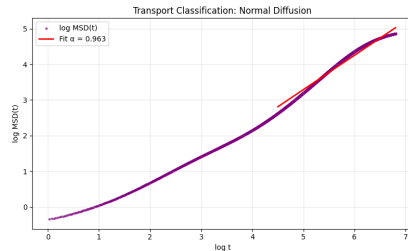
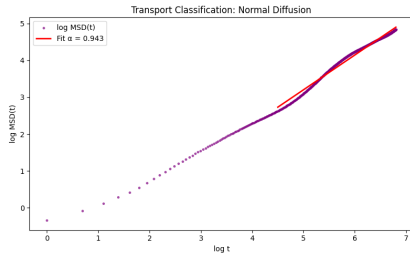
PINN vs FDM concentration evolution for two absorbing boundaries.

Transport Metrics and Survival Probability

PINN Results

$$\alpha_{\text{PINN}} = 0.943$$

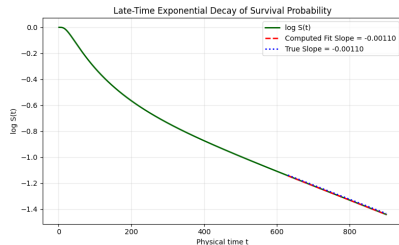
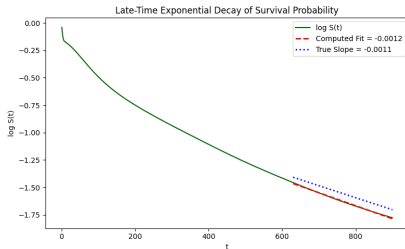
$$\text{slope} = -0.0012$$



FDM Results

$$\alpha_{\text{FDM}} = 0.963$$

$$\text{slope} = -0.00110$$



Both closely match theoretical exponential decay. However, FDM performed more accurately

2D Bounded Diffusion

Governing Equation:

$$u_t = D(u_{xx} + u_{yy}), \quad D = 0.1, \quad x, y \in [-5, 5], \quad t \in [0, 50]$$

Boundary Conditions

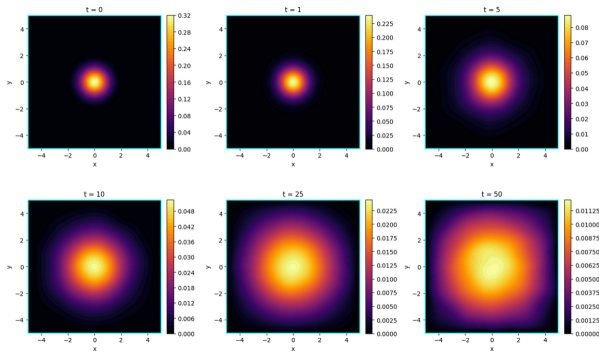
Four perfectly absorbing walls.

Collocation Point Strategy

- Total collocation points: 21,000
- Interior points: 15,000
- Boundary points: 2,000
- Initial-condition points: 4,000

Observation

As dimensionality increases, PINNs exhibit higher computational error and training cost compared to FDM.



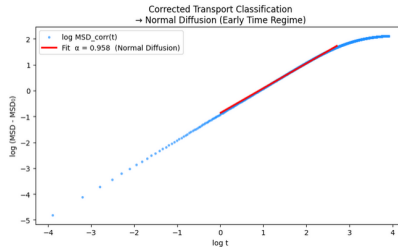
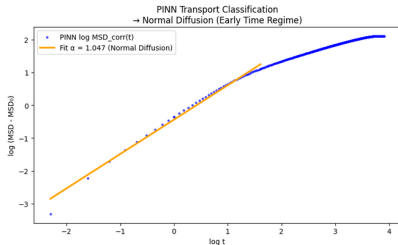
2D diffusion snapshots showing circular spread constrained by square absorbing boundaries.

2D Transport Metrics

PINN Results

$$\alpha_{\text{PINN}} = 1.047$$

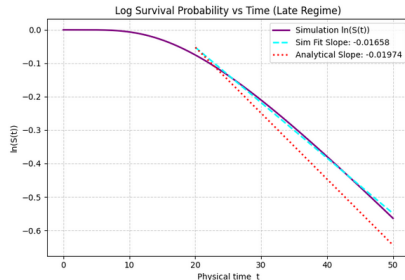
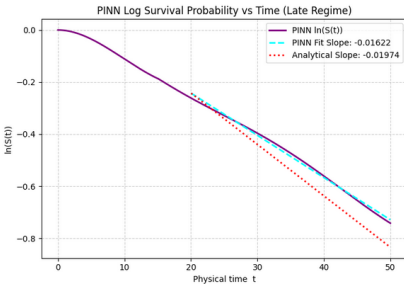
$$\text{slope} = -0.01622$$



FDM Results

$$\alpha_{\text{FDM}} = 0.958$$

$$\text{slope} = -0.01658$$



Inverse Problem: Hidden Defect Discovery

Governing Equation:

$$u_t = Du_{xx} - S(x)u, \quad D = 0.5, \quad L = 10$$

Problem Objective

Discover an unknown hidden defect profile purely from sparse and noisy sensor measurements.

Hidden Defect

$$S(x) = 2.0$$

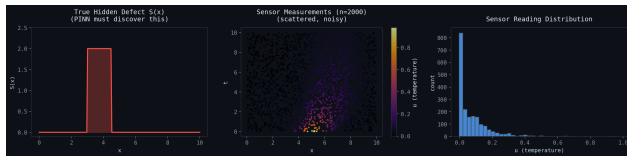
embedded within: $x \in [3.0, 4.5]$

Sensor Data

- 2000 scattered measurements
- Generated using Explicit FDM
- Gaussian noise added

Key PINN Advantage:

Classical FDM solves only forward problems, while PINNs can simultaneously discover unknown physical parameters directly from observed data.



Ground-truth hidden defect and noisy sensor measurements.

DualPINN Architecture

• Temperature Network

- 6 hidden layers
- 128 neurons
- Predicts $u(x, t)$

• Defect Network

- 4 hidden layers
- 64 neurons
- Predicts $S(x)$
- Softplus activation ensures $S(x) \geq 0$

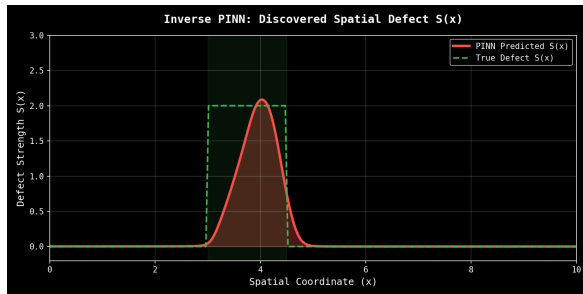
Optimization Strategy

- Adam optimizer
- 20,000 epochs
- Staged physics-loss weighting

Results

$$x_{\text{detected}} \in [3.07, 4.76] \quad S_{\text{max}} = 2.0861$$

This inverse problem demonstrates one of the strongest advantages of PINNs over traditional FDM methods.



PINN predicted defect profile vs hidden ground truth.

1. FDM for Standard Forward Problems

- Exceptional computational speed
- Deterministic numerical accuracy
- Reliable transport metric estimation

2. PINNs for Complex Boundaries

- Mesh-free framework
- Naturally handles mixed BCs
- No ghost-node approximations
- No CFL constraint

3. PINNs for Inverse Problems

- Unknown parameters become trainable variables
- Hidden defect successfully recovered from noisy data

4. Nature of Error

- FDM: cumulative truncation error
- PINNs: optimization error with front-loaded training cost

PINNs and FDM are complementary computational tools.

Thank You